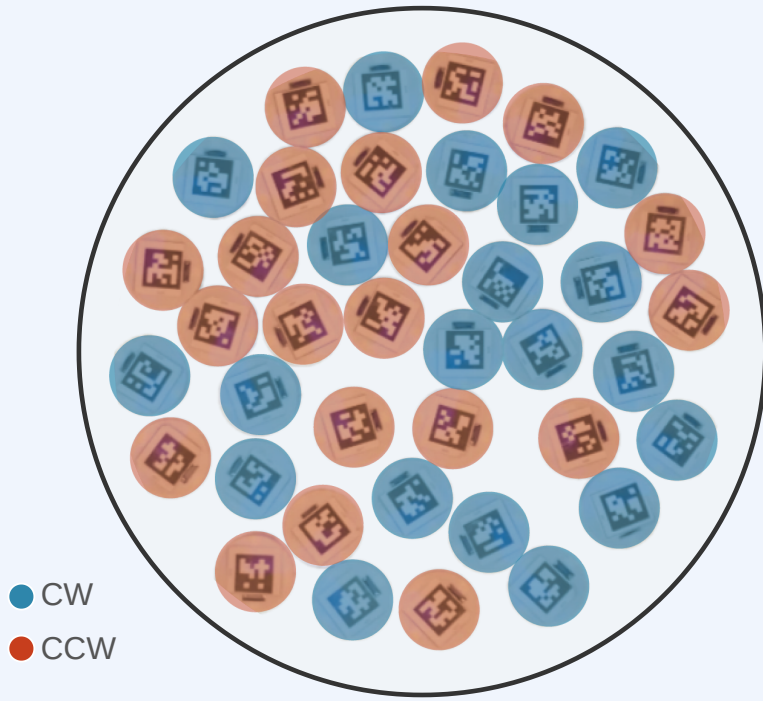


MICROSCOPIC

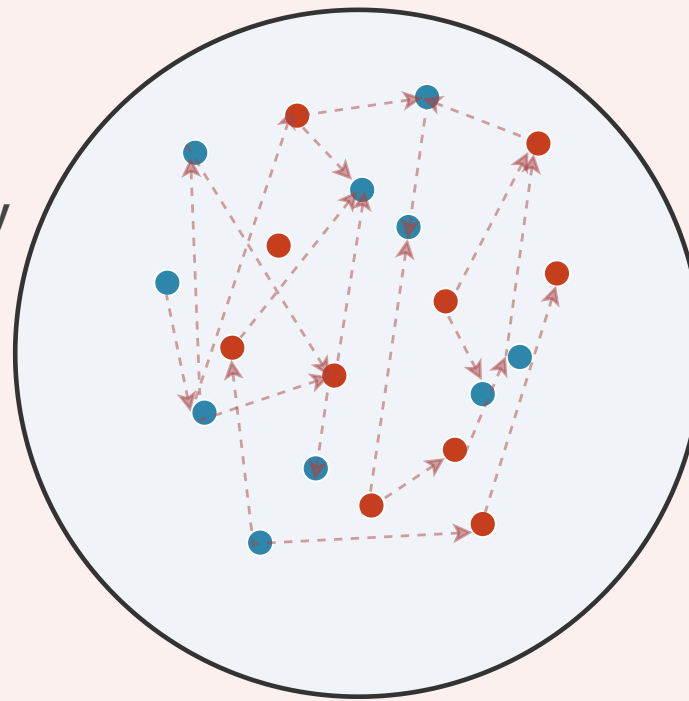


Bristle-Bot Swarm
40 self-rotating robots
(20 CW + 20 CCW)

ODE Discovery

$$\ddot{r}_k = F_{\text{int}} + \Sigma F_{\text{ext}}$$

MESOSCOPIC

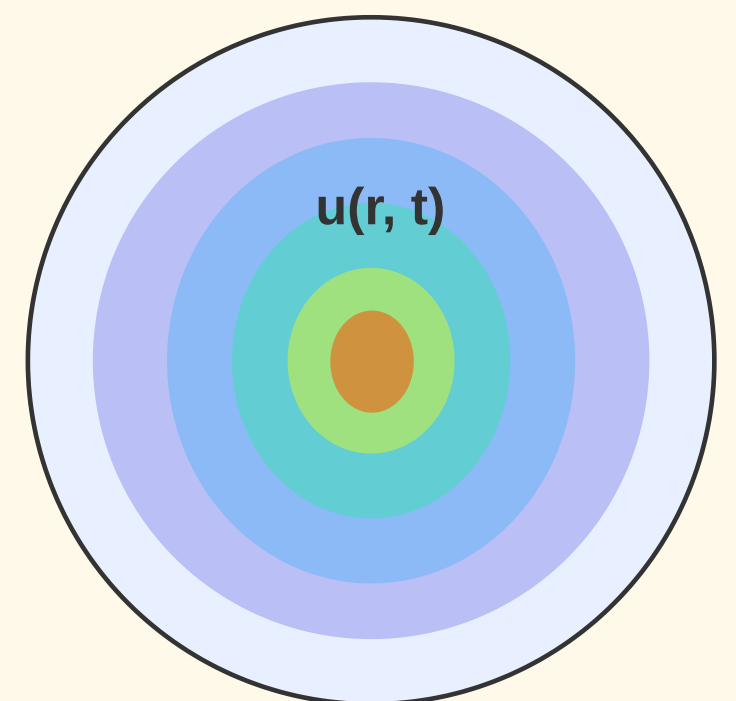


Order Parameters
 $\Phi(t) = [P, V_{\text{cm}}, L]$
5-robot subsystem

SDE Discovery

$$d\Phi = A dt + B \circ dW_t$$

MACROSCOPIC



Continuous Velocity Field
Reconstructed via kriging

SPDE Discovery

$$\partial_t u = \mathcal{L}_\mu[u] + \mathcal{L}_\sigma[u] \xi$$